CYBERSECURITY PROJECT

PROJECT DISCOVERY

CYBER DIARIES

Based on the OWASP Top 10:2025

An Interactive Arabic Platform for

Learning Web Application Security



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THE PROBLEM

Why learners struggle

- Theory and practical skills are often separated.
- Learning resources are scattered across different websites.
- English-heavy materials can be difficult for Arabic-speaking learners.
- Exercises may not reflect real workplace situations.
- Progress is hard to measure beyond finishing lessons.



OUR PROPOSED SOLUTION

One connected Arabic learning experience

- Storytelling connects learning activities.
- Arabic explanations introduce security concepts.
- Videos and external resources support understanding.
- Isolated labs provide hands-on practice.
- Progress tracking helps learners understand their development.



FIRST RELEASE SCOPE

One complete episode: A05:2025 — Injection

- Story scene and technical explanation.
- Video resource or backup link.
- Short quiz with 3–5 multiple-choice questions.
- Practical lab, story introduction, role and difficulty selection.
- Automatic verification, feedback, summary and optional diary.
- Track episode progress and completion time.



TARGET LEARNERS

Built for Arabic-speaking learners

- Students connecting academic knowledge with practical skills.
- Beginners learning web application security.
- Developers improving their understanding of vulnerabilities.
- Security professionals exploring incident-handling roles.
- Arabic explanations with key English technical terms.



THE LEARNING JOURNEY

A complete cycle for each incident

- DISCOVER — identify the issue in the authorized training scenario.
- INVESTIGATE — analyze evidence and understand what happened.
- FIX — address the root cause and validate the change.
- DOCUMENT — record actions, findings and lessons learned.
- The stages connect knowledge to practical work.



ONE INCIDENT, THREE ROLES

Different responsibilities, shared scenario

- RED TEAM — discover and safely demonstrate the vulnerability.
- BLUE TEAM / SOC — analyze evidence, logs and affected resources.
- DEVELOPER — identify the root cause, fix it and validate the result.
- Each role has distinct tasks within the same incident.
- Learners can explore one role or complete the full cycle.



DIFFICULTY, FEEDBACK & POINTS

Guidance changes; the learning goal stays the same

- EASY — broad hints.
- MEDIUM — directional hints.
- HARD — no hints.
- Difficulty changes guidance, not the solution.
- Points encourage practice and are awarded for verified success.
- Points never unlock or block learning content.



SAFE, ISOLATED LABS

Safety is a core design requirement

- Each exercise runs in an isolated container.
- Network access is restricted to protect real systems.
- Privileges are limited to what the exercise needs.
- Containers are destroyed when sessions end or expire.
- Browser-based access; no tools installed on the learner’s device.
- Targets: 60-minute default, 120-minute maximum, 10 concurrent learners.



PROPOSED TECHNOLOGY STACK

Modular architecture with a separate lab service

- FRONTEND — Next.js 14.
- BACKEND — ASP.NET Core / .NET 10 with REST APIs.
- DATABASE — PostgreSQL 16.
- LAB INFRASTRUCTURE — Docker containers.
- MAIN APPLICATION — Modular Monolith.
- The lab service remains separate from the main API.



PROGRESS & LEARNING REFLECTION

Measure real activity, not just time online

- Track lab sessions, completed tasks and episode completion.
- Measure completion ratios and time spent.
- Optional diary: What happened? Why?
- What did I do? What did I learn?
- Skipping the diary does not block progress or reduce points.



RESEARCH & EVALUATION

Evaluate learning and usability

- Pre-test to measure knowledge before the episode.
- Post-test to measure knowledge after the episode.
- Delayed test if feasible.
- Analyze task completion, time and user feedback.
- Initial pilot: approximately 5–10 users.
- Treat results as exploratory, not strong statistical proof.



PRIVACY & RESPONSIBLE USE

Protect learners and keep practice ethical

- Collect only the data needed for learning and evaluation.
- Keep personal information and secrets out of educational logs.
- Define retention rules and an account-deletion process.
- Obtain informed consent and required university ethics approval.
- Use fictional characters and isolated labs for defensive learning.



CURRENT STATUS & KEY RISKS

What needs attention before full implementation

- Complete missing functional requirements and acceptance criteria.
- Finalize usability and non-functional requirements.
- Resolve privacy, research and tool-licensing decisions.
- Validate lab isolation early.
- Risks include scope expansion, difficult automated checks and weak role-specific tasks.
- Priority: prove lab safety and feasibility first.



ROADMAP & NEXT STEPS

A focused, staged implementation plan

- PHASE 1 — complete requirements and open decisions.
- PHASE 2 — build and test the isolated Docker lab prototype.
- PHASE 3 — develop the platform and Injection episode.
- PHASE 4 — pilot with 5–10 users and improve.
- Do not expand scope before measuring the first complete episode.



CONCLUSION

A practical and accessible way to learn web security

- Arabic learning materials and connected storytelling.
- Safe practical labs and three professional perspectives.
- A complete cycle: discover, investigate, fix and document.
- Progress tracking and research-based evaluation.
- Immediate goal: deliver and evaluate one complete Injection episode.

